



Exploring the World of Science

2025 Hawk and Hornet Invitational

Machines Division C

November 15, 2025



Instructions:

- Tiebreakers 1st – Best ES, 2nd – Best TS, 3rd – Best R1, 4th – Questions 9, 11, 12, 19, 28, 33
- Each team's ES score, out of 80.5 points, will be scaled to be out of 50 points
- Use $g = 9.8 \frac{m}{s^2}$ unless otherwise noted
- Round values to 3 decimal places
- There are 9 sections to this test
- This test was written by Lucia Lee (Athens High School Alum '22)
- Lab portion will be during the test
- A wise teacher once said, RIF, Reading is Fundamental
- Capital letters indicate a multi-part question, lowercase letters are multiple choice
- Good Luck!

Team Name: _____

Team Number: _____

Section 1. General

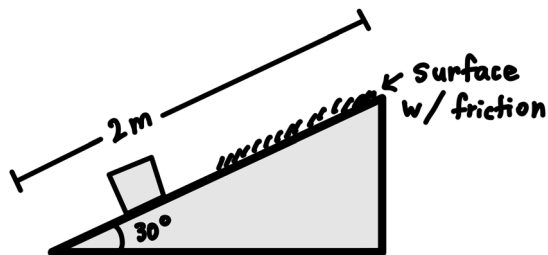
- 1) Which of the following is the most comprehensive definition of a simple machine? (2 points)
 - a. An elementary device that has a specific movement, which can be combined with other devices and movements to form a complicated machine.
 - b. A non-electronic device created for the purpose of comparing weights of objects or make physical jobs easier, by changing the magnitude or direction of the force.
 - c. Mechanical devices combined together to apply force to reduce the work needed in order to accomplish a task.
 - d. A device that modifies motion and force to perform work
- 2) List five types of simple machines and for two, explain how each can make work easier (3 points – 1 point for list five simple machines, 1 point per each of the two explanations):
- 3) Write the mass of a 10 kg box in terms of *fundamental* SI units (1 point).
- 4) Which of the following are composed of a *single* simple machine (Mark all correct answers)? (2 points)
 - a. Blade of a Pencil Sharpener
 - b. Fishing Rod
 - c. Nail Clipper
 - d. Screwdriver
 - e. Roller Coaster
- 5) Write the name of this device based on its description and provide identify the type of simple machine it is (3 points):
 - A. Consists of a horizontal cylinder (barrel), which is rotated by the turn of a crank or belt. A winch is affixed to one or both ends, and a cable or rope is wound around the winch, pulling a weight attached to the opposite end.
 - B. Used for lifting water from a low-lying body of water into irrigation ditches, it lifts water by turning a spiral-shaped surface inside a pipe.
 - C. This machine is operated by pulling upon the slack section of a continuous chain that wraps around two pulleys on a common shaft. (The two pulleys are joined together such that they rotate as a unit on the single shaft that they share.) It is used to manually lift very heavy objects like car engines.

If applicable, provide the standard dimensions for the following. Dimensions are types of measurements, such as time and length. The dimension of speed is $\frac{\text{length}}{\text{time}}$.

- 6) Provide the dimensions of Energy (1 point)
- 7) Provide the dimensions of Momentum (1 point)
- 8) Provide the dimensions of Efficiency (1 point)

Section 2. Newton

- 9) A 5 kg block of mass moves across a plane 2 meters long with an incline of 30° at 2 m/s. The first meter of the plane is a smooth, frictionless surface, while the second meter requires 20% more work due to friction for the object to maintain the same velocity. What is the coefficient of friction of the second half of the plane's surface? (4 points)



- 10) Match the following phenomena to Newton's first, second or third law of motion.
 - A. It takes more force to push a wagon with a child than an empty wagon. (1 point)
 - B. Pins hit by a bowling ball are knocked down. (1 point)

C. A swimmer's stroke pushes on the water and they move forward. (1 point)

11) You are standing at a roller rink with inline skates and are (dangerously) holding a bowling ball. You decide to throw the ball at $10.0 \frac{m}{s}$ at an angle of 15 degrees above the horizontal. If you are 65.0 kg and the ball has a mass of 4.54 kg, what is your speed after you throw the bowling ball (surface is frictionless)? (4 points)

12) A wire is bent into a square, with a side length of 25 cm. If the wire's total mass is 3.1 g, find the moment of inertia about an axis through the center and perpendicular to the plane of the square. Please provide an answer in standard units. (note: The wire is a thin uniform rod; hint: Use parallel axis theorem) (4 points)

13) Calculate the IMA of a truck shown below (length 5.36 feet, height 3.4 feet) (3 points)



- 14) What is the AMA of the truck ramp from question 13 if the ramp has frictional constant of .18? (3 points)
- 15) How much work is needed to push a crate of 83 kg up this truck ramp given friction still exists? (Note: 1 meter = 3.28 foot) (2 points)
- 16) The mechanical advantage of an inclined plane is the _____ of the slope divided by the _____ of the slope. (2 points)
- 17) Why is the efficiency of a simple machine always less than 100%? (2 points)
- 18) At what instance is the AMA equal to its IMA? (2 points)
- 19) A cube with equally distributed mass of 0.7 kg is placed in a parallel universe where smooth frictionless ramps exist. As it is traveling down at .8 m/s, suddenly, the cube is instantaneously transported to an equivalent ramp on earth where it continues sliding down, now, with a friction constant of 0.52. If the incline of the mountain is 27 degrees, how long will the block have traveled on the surface with friction until it comes to a stop? (4 points)

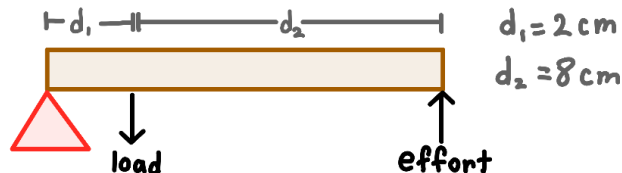
- 20) A block's mass is 50 kg and it has been pushed down an inclined plane so that it is traveling at 100 m/s. If this inclined plane has the same friction constant, but is infinitely long, will the block still come to a stop? Explain why or why not. (2 points)

Section 3. Levers

- 21) Identify which lever category each of the following objects are (8 points – 0.5 points per correct object)

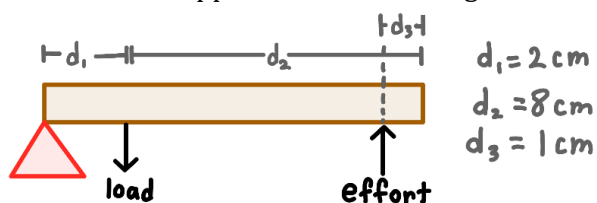
- Brake pedal of a car
- Crowbar
- Wheelbarrow
- Hammer
- Nutcracker
- Bottle Opener
- Scissors
- Pair of tweezers

- 22) Given that the actual mechanical advantage of the lever is 4, calculate the efficiency. (2 points)



- 80%
- 75%
- 50%
- 40%
- 25%

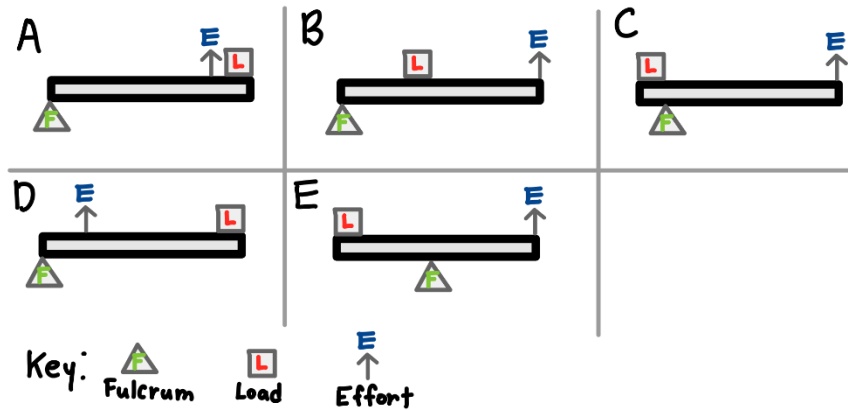
- 23) Given the weight, evenly distributed across the lever, is 1 kg, calculate the effort force needed to support a load of 3 kg as it is depicted below. (2 points)



- 0.87 N
- 6.5 N
- 7.1 N
- 8.5 N
- 11.9 N

24) The mechanical advantage of a lever is the length of the _____ arm divided by the length of the _____ arm. (2 points)

25) Of the five levers below, which has the highest IMA? Please circle the letter of the correct answer. (2 points)



Section 4: Wedges

26) True or False: Wedges depend on the friction between the outer surface and the region it is pushing against. (1 point)

27) A lumberjack swings a 3-kg axe head at 15 m/s to split a log. If the axe head decelerates in 0.1 seconds during the collision, what force does the log impart on the axe? (3 points)

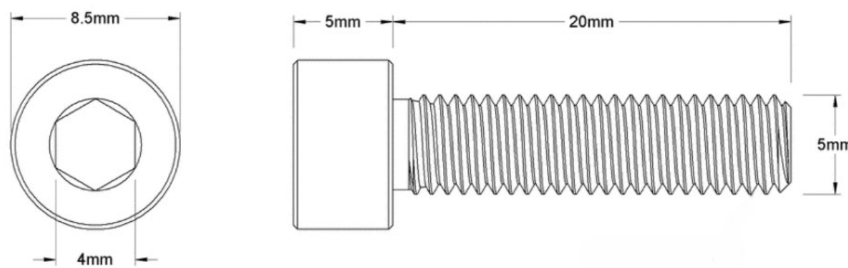
28) Various kitchen knives are used around the household, including the bread knife (image on left) and the chef's knife (image on right). Though they are both wedges, they are shaped differently. Explain why the bread knife uses a back-and-forth sawing motion while the chef's knife uses a single slicing motion from the tip of the blade. (2 points)

(Note the toothed blade on the edge of the bread knife)



Section 5: Screws

29) The following is a diagram of a stainless steel socket head screw from a well-known manufacturer and distributor. The pitch is 0.8 mm.



- A. This screw is a M_ x _ mm thread screw (1 point)
- B. What is the IMA of the screw if effort is applied by a 10 cm long wrench? (2 points)

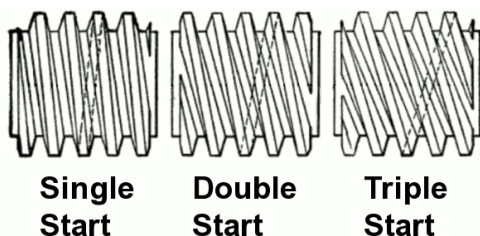
30) In screw assemblies, one must consider the possibility of back driving. Why might a greater efficiency lead to a greater tendency for the screw to back drive? (1 point)

31) What is something that will *not* strengthen the connection of a screw? (1 point)

- Wider diameter
- Longer screw length
- Heat treatment
- Coarser threads

32) The following questions will be about the lead and pitch of a screw.

- A. What is a lead and pitch of a screw? (1 point)
- B. Label the lead and pitch of each of the starts on the diagram below (1 point)



Section 6. Gears

33) Identify each of the gears and indicate if the following is a correct use case for the gear. There will be one remaining image. (2.5 points – 0.5 per correct match)

- A. Rack and Pinion
- B. Helical Gear
- C. Hypoid Gear
- D. Worm Gear
- E. Miter Gear

